

# Timer Replacement PCB — Complete Documentation

**Translation.** This is the English translation of the manual. The authoritative version is the German original ([Deutsch](#)). Also available: [Français](#).

**Project:** Replacement of a defective/outdated timer control PCB with an in-house development using an ESP32-C3, OLED display, Wi-Fi configuration and a Shelly 1PM Mini Gen4 as the power switch.

**Status/Date:** 2026-07-13 · Template version v8 · Phase: verification before layout **Author:** Silvio (hardware/measurements) + Claude (analysis/design)

## 1. What the device does (functional description)

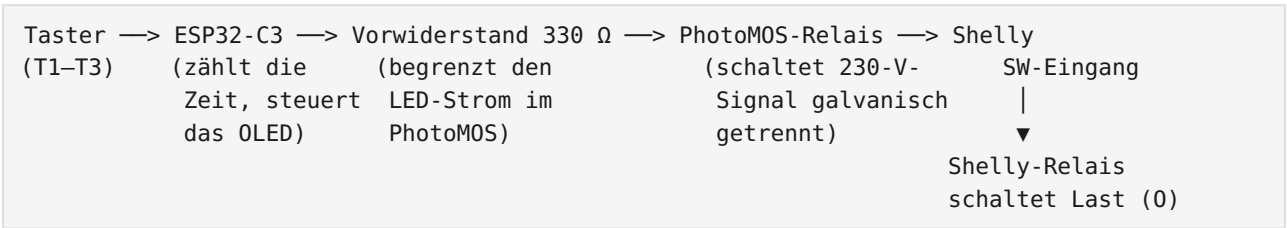
The device has three buttons on the front. Pressing a button switches a 230 V load on for an adjustable time:

| Button | Default time | adjustable        |
|--------|--------------|-------------------|
| T1     | 5 seconds    | 1-600 s via Wi-Fi |
| T2     | 10 seconds   | 1-600 s via Wi-Fi |
| T3     | 15 seconds   | 1-600 s via Wi-Fi |

A small OLED display behind the viewing window shows the **Wi-Fi reception** (left) at the top, the large **time of day** (NTP, center) and the status "Ruhe" (Idle, right); at the bottom left the **date** (e.g. "Do 16.07.2026") and at the bottom right the **free memory**. While a timer is running, the center shows the **seconds countdown** and the right shows "Futter" (Feed). The three buttons (**S1 Down/Manual**, **S2 SET**, **S3 UP**) trigger timers, stop them and page through an info menu — details in Ch. 6.5.

Operation runs **without an app and without a cloud** via a small web page optimized for mobile devices, which the ESP32 serves itself on **port 80** (open `http://feeder-relais.local` in the browser): trigger buttons, set times, view network and status values. For integration (e.g. with ioBroker) the ESP additionally offers a lean **JSON API** (see Ch. 6.4). Optionally, the device can also be integrated into Home Assistant or ioBroker (esphome adapter).

### 1.1 The signal chain (this is how the signal flows)



**Why a Shelly?** The Shelly 1PM Mini Gen4 handles the actual switching of the load, measures the power in the process, and is additionally reachable via Wi-Fi/app. Our board only "presses" the switch for the Shelly — as a real hardware signal, **without a Wi-Fi dependency** for the core function.

**What is a PhotoMOS?** A tiny solid-state relay: on one side an LED (driven by the ESP with 3.3 V), on the other side a light-controlled switch that is allowed to switch mains voltage. Between them there is only light — i.e. a complete galvanic isolation between low voltage (SELV) and 230 V. The Shelly's SW input is mains-referenced (it expects a switched L), which is why a type with  $\geq 400\text{ V}$  **reverse voltage is mandatory** (e.g. Omron G3VM-601BY or Panasonic AQY216). An ordinary optocoupler (PC817) is **not** permissible here.

The Shelly is configured to input mode "**Switch/Follow**": its relay is on for exactly as long as the ESP holds the signal. The timing is entirely up to the ESP.

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## 2. Safety — read this first

- The device works with **230 V mains voltage. Danger to life!** Assembly, testing and commissioning belong in the hands of a suitably qualified person (a qualified electrician or an instructed person).
- Initial functional tests of the ESP, buttons and OLED are performed **exclusively over USB** — with no mains voltage connected whatsoever (see the test plan in the project plan).
- The first 230 V test is performed behind an **RCD/PRCD (portable residual current device plug)**, with the enclosure closed; never work on the open device while it is powered.
- On the board the rule is: at least **6 mm of distance** (creepage distance) between all 230 V nets and the low voltage (SELV); details in Ch. 7.6.
- A fuse (1 A slow-blow) and a varistor upstream of the power supply module are mandatory — the modules (TSP-05/HLK) do not bring either of them along themselves.

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## 3. Bill of materials (BOM)

| Ref   | Component            | Value / type                                | Note                                                                |
|-------|----------------------|---------------------------------------------|---------------------------------------------------------------------|
| U1    | ESP32-C3 Super Mini  | 18 × 24 mm, USB-C                           | controller, Wi-Fi                                                   |
| U2    | PhotoMOS relay       | Omron G3VM-601BY <b>or</b> Panasonic AQY216 | $\geq 400\text{ V}$ ! SOP-4/DIP-4                                   |
| PS1   | AC/DC module         | TENSTAR TSP-05 (5 V / 3 W)                  | HLK-PM05 clone, 34.7 × 20.5 × 15.05 mm, <b>measure pin spacing!</b> |
| —     | Shelly 1PM Mini Gen4 | S4SW-001P8EU                                | 29 × 34 × 16 mm, max. 8 A/240 V                                     |
| —     | OLED display         | 0.91" SSD1306, 128 × 32, I2C                | module 38 × 12 mm, pins: GND VCC SCL SDA, address 0x3C              |
| SW1-3 | Short-travel button  | 6 × 6 mm, travel 1.5 mm                     | pin pattern 6.9 × 4.4 mm (measured from the original)               |
| R1    | Resistor             | 330 $\Omega$ , axial                        | LED series resistor for PhotoMOS                                    |
| F1    | Fine-wire fuse       | 1 A slow-blow, 5 × 20 mm + holder           | primary side                                                        |
| RV1   | Varistor (MOV)       | S14K275                                     | primary side, in parallel                                           |
| C1    | Electrolytic cap     | 220 $\mu\text{F}$ / 10 V                    | 5 V buffer for Wi-Fi current peaks                                  |
| C2    | Ceramic cap          | 100 nF                                      | 5 V decoupling                                                      |
| J1    | Mains terminal       | 2-pole (L, N)                               | screw terminal or solder pads                                       |
| J2    | Shelly connection    | 4 solder pads/pins                          |                                                                     |

| Ref | Component         | Value / type                    | Note                                                    |
|-----|-------------------|---------------------------------|---------------------------------------------------------|
|     |                   |                                 | L, N, SW, O — short wires to the Shelly screw terminals |
| J3  | Load output       | 2-pole                          | O (switched), N                                         |
| J4  | OLED socket       | socket header 1 × 4, pitch 2.54 | build height per stack measurement (~11 mm), Ch. 5.6    |
| —   | Pin header 1 × 4  | pitch 2.54                      | soldered to the OLED (facing backward)                  |
| —   | Mounting material | 3M VHB adhesive tape or 3D clip | fastening the Shelly + TSP                              |

## 4. Enclosure and mechanics

### 4.1 The coordinate system (important for everything that follows!)

All dimensions refer to the **upper left corner of the ORIGINAL PCB**, x to the right, **y downward**, viewed from the **component side** (buttons at the top). The new, larger board extends beyond this origin — which is why there are **negative coordinates**. Advantage: all values ever measured remain valid forever, no matter how the contour grows.

### 4.2 Boundary conditions from the enclosure (measured/derived)

| Quantity                        | Value                                         | Source                                                                       |
|---------------------------------|-----------------------------------------------|------------------------------------------------------------------------------|
| Inner overall height            | 19.9 mm                                       | measurement Silvio                                                           |
| Frame height (front panel)      | 2 mm                                          | measurement Silvio                                                           |
| Space <b>above</b> the board    | <b>16.5 mm</b>                                | measurement Silvio (board ↔ front)                                           |
| Board thickness                 | 1.6 mm                                        | standard                                                                     |
| Space <b>below</b> the board    | <b>≈ 1.8 mm</b>                               | 19.9 – 16.5 – 1.6 → nothing may protrude downward!                           |
| Height limit in the window area | <b>≈ 12 mm</b>                                | glass + OLED plugged in front of it                                          |
| Board fastening                 | 4 screws on 16.5 mm bosses on the front panel | with stabilizing ribs to the edge                                            |
| Enclosure screwing              | 6 external screws                             | the board has 6 edge-open U-slots (width 10) instead of closed feed-throughs |

### 4.3 Final board geometry (template v9)

All values are stored **machine-readable** in `hardware/platine_template.py` (the single source of truth!). Summary:

| Element                  | Coordinates (mm)                                                                                                                              | Accuracy       |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Contour                  | x –14.2...91.1 / y –11.8...67.6 → <b>105.3 × 79.4</b> , corner radius 4 (locally reduced to 3.2 and 2.1 respectively at the outer boss slots) | ±1 (left ±1.5) |
| 4 mounting holes<br>ø3.0 | (3.8/4.1) (73.1/4.0) (3.7/39.6) (73.4/50.7)                                                                                                   | ±0.3           |

| Element                              | Coordinates (mm)                                                                                                                                        | Accuracy                                                                            |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 6 boss slots (edge-open U, width 10) | upper row y −6.0 (opens toward the top edge), lower row y +64.0 (opens toward the bottom edge), row spacing 70; columns x −6.0 / 39.0 / 84.0 (pitch 45) | measured directly (Silvio 13.07.); absolute position still to be <b>test-fitted</b> |
| Buttons T1/T2/T3 (centers)           | (13.2/12.3) (38.8/12.2) (64.1/12.0)                                                                                                                     | ±0.3                                                                                |
| Viewing window (reference)           | x 18.4...53.6 / y 23.3...40.0                                                                                                                           | ±1                                                                                  |
| J4 OLED socket (4 pins vertical)     | pins at x = 15.0, y = 27.84 / 30.38 / 32.92 / 35.46                                                                                                     | check assumptions, Ch. 5.6                                                          |
| Mounting area Shelly                 | x 56...90 / y 16.5...45.5 (horizontal)                                                                                                                  | —                                                                                   |
| Mounting area ESP32-C3               | x 8.5...32.5 / y −10...8 (transverse, USB upward)                                                                                                       | —                                                                                   |
| Mounting area TSP-05                 | x −1.5...33.2 / y 44.2...64.7 (AC side to the left)                                                                                                     | —                                                                                   |
| Keepouts (reference)                 | ø8 around the 4 screw heads + 5 rib bands                                                                                                               | rib directions = assumption!                                                        |

**Where do the values come from?** Solder-side scan of the original PCB (300 dpi, de-rotated, mirrored) for buttons/holes/contour; enclosure scan, registered via the 4 frame screws (mirror transformation  $x_{\text{platine}} = 110,5 - x_{\text{gehäuse}}$ ,  $y_{\text{platine}} = y_{\text{gehäuse}} - 16,4$ , ±0.4 mm); window position from the perspective-corrected photo.

## 4.4 The old cutout

The trapezoidal cutout on the bottom edge of the original PCB was the clearance for the middle lower screw boss (B5). In the new board it is **replaced by the edge-open U-slot B5** and is omitted.

# 5. Electronics in detail

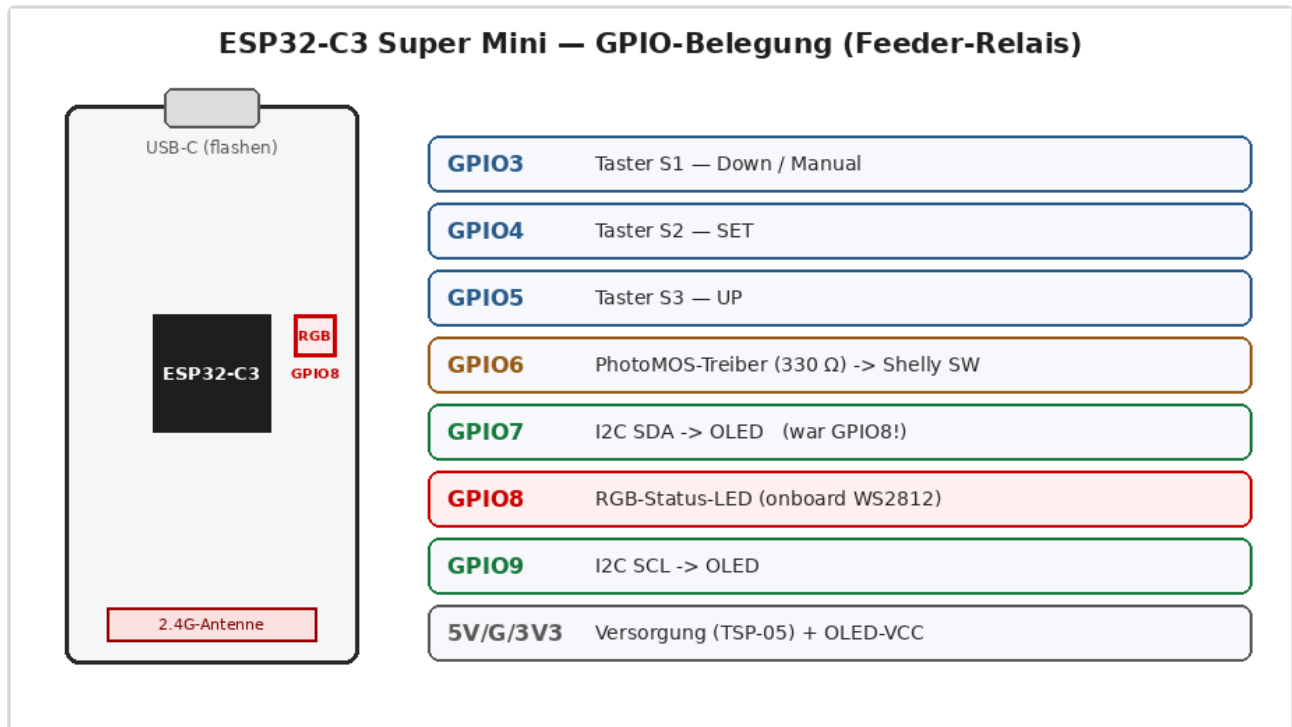
## 5.1 Overview of the nets (connections)

| Net name          | carries            | connects                                          |
|-------------------|--------------------|---------------------------------------------------|
| L_IN              | 230 V              | mains terminal J1.L → fuse F1                     |
| L_F               | 230 V (fused)      | F1 → varistor, TSP-05 AC, PhotoMOS OUT1, Shelly L |
| N                 | 230 V neutral      | J1.N → varistor, TSP-05 AC, Shelly N, load J3     |
| SW_SHELLY         | 230 V switched     | PhotoMOS OUT2 → Shelly SW                         |
| O_LAST            | 230 V switched     | Shelly O → load output J3                         |
| +5V               | low voltage (SELV) | TSP-05 +Vo → ESP 5V, C1, C2                       |
| GND               | low voltage (SELV) | TSP-05 −Vo → ESP GND, buttons, PhotoMOS K, OLED   |
| +3V3              | low voltage (SELV) | ESP 3V3 output → OLED VCC                         |
| BTN1/2/3          | signal             | buttons → GPIO3/4/5 (internal pull-ups)           |
| PMOS_DRV → PMOS_A | signal             | GPIO6 → R1 330 Ω → PhotoMOS anode                 |
| SDA / SCL         | I2C                | GPIO7 / GPIO9 → OLED                              |

| Net name | carries | connects                                    |
|----------|---------|---------------------------------------------|
| RGB-LED  | signal  | GPIO8 → onboard WS2812 (status LED, dimmed) |

## 5.2 ESP32-C3 Super Mini: board and pin assignment

Module variant "ESP32-C3FN4". An overview of the pins we use:



Components on the module: **1** USB Type-C · **2** BOOT button · **3** RESET button · **4** LDO CAT6219 (3.3 V, 500 mA) · **5** ESP32-C3FN4 · **6** 2.4 GHz antenna · **7 RGB module (WS2812, GPIO8)** · **8** antenna connector (u.FL).

| Pin    | Function                  | Note                                                             |
|--------|---------------------------|------------------------------------------------------------------|
| 5V / G | supply                    | from the TSP-05                                                  |
| 3V3    | output of the onboard LDO | supplies the OLED                                                |
| GPIO3  | button T1                 | pull-up internal, button to GND                                  |
| GPIO4  | button T2                 | ditto                                                            |
| GPIO5  | button T3                 | ditto                                                            |
| GPIO6  | PhotoMOS driver           | high = Shelly on                                                 |
| GPIO7  | SDA (I2C)                 | OLED — <b>instead of GPIO8</b> , which is where the RGB LED sits |
| GPIO8  | RGB LED (WS2812)          | onboard status LED, dimmed                                       |
| GPIO9  | SCL (I2C)                 | OLED                                                             |

Deliberately avoided: GPIO2/8/9 as buttons (strapping pins). **GPIO8 carries the onboard RGB LED (WS2812)** — which is why I2C **SDA is on GPIO7** (not GPIO8), otherwise the I2C traffic would drive the LED brightly (heat). SCL stays on GPIO9 (boot-compatible with an I2C pull-up).

## 5.3 Shelly 1PM Mini Gen4

- Terminals: **SW** (switch input), **O** (load output), **L**, **N**.

- Supply 110–240 V~, relay max. 8 A / 2000 W @ 240 V.
- Configuration after commissioning: Input Mode = Switch (Follow), so that the relay follows the SW signal 1:1.
- Mounting: lying flat on the mounting area, VHB adhesive tape or 3D clip. Four short wires (e.g. 0.75 mm<sup>2</sup>, 6–7 mm stripped) from the screw terminals to the J2 pads.

## 5.4 Power supply TSP-05

100–240 V~ → 5 V DC / 3 W (= 600 mA). Sufficient for the ESP Wi-Fi peaks (~350 mA) plus the OLED with reserve. **Open item: check the pin spacing with calipers against the KiCad footprint Converter\_ACDC\_HiLink\_HLK-PMxx** (the clone is usually pin-compatible — but at 230 V, assume nothing).

## 5.5 Buttons

The three 6×6 short-travel buttons (1.5 mm travel) are actuated by plungers in the front panel — their position is therefore **non-negotiable** (place them to within ±0.2 mm). Pin pattern per button: 6.9 mm × 4.4 mm.

## 5.6 OLED — plug-in mounting ("stack")

The OLED is **not soldered**, but **plugged onto J4**: socket header 1×4 on the board, pin header on the OLED facing backward, the display face rests against the glass pane (with a thin foam pad).

- Assignment of J4 from top to bottom: **GND, VCC (3V3), SCL, SDA** — identical to the labeling of the existing module.
- **Required stack height** = 16.5 mm – glass inset – module thickness ( $\approx 2.8$  mm)  $\approx$  **13–14 mm**. Realized for example with an 11 mm socket header + 2.5 mm pin header collar. Socket headers are available in 8.5/11/13/15 mm.
- **Measure on the real module before freezing** (manufacturers vary by ±1 mm): 1. distance module left edge → pin row (assumption 2.5 mm), 2. distance module left edge → center of the active area (assumption 23.5 mm), 3. glass inset (front inner face → underside of glass). In practice: lay the module in the frame, display centered in the window, measure the pin holes relative to two boss screws → that is J4.

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## 6. Firmware (ESPHome)

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File: `firmware/timer-relais-c3.yaml` (device name **feeder-relais**, mDNS `feeder-relais.local`). Contains everything: Wi-Fi with fallback hotspot ("Feeder-Relais Setup", captive portal), **NTP clock** (`de.pool.ntp.org`), mobile web app + JSON API (Ch. 6.4), the three persistent times, button logic with debounce/long press/menu (Ch. 6.5) and an OLED with Wi-Fi status, clock or countdown.

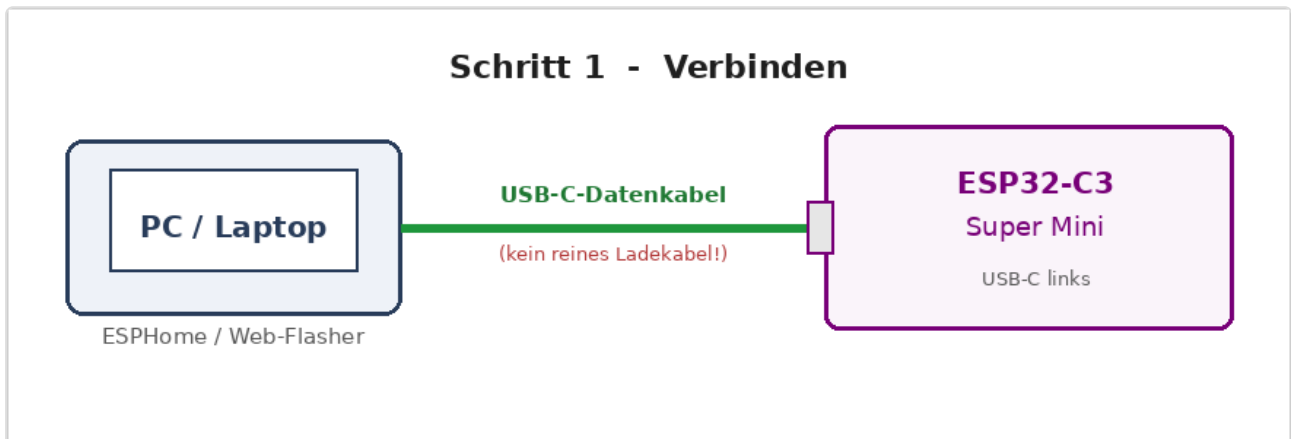
### 6.1 First flashing — step by step (for beginners)

The **very first time**, the firmware is put onto the ESP32-C3 **via a USB cable**. All later updates then run wirelessly (Ch. 6.6).

**This is what you need:**

- the **ESP32-C3 Super Mini**,
- a **USB-C data cable** — caution: many cheap cables can only charge; with those the chip is **not detected**!

- a PC with **Google Chrome** or **Microsoft Edge** (browser route B) — or ESPHome on the PC (route A).



**Ready-made images are available** — so you don't have to compile anything yourself:

- in the repo under `firmware/build/` (always the current state):
- `feeder-relais.factory.bin` — full image for the **first flash** (address 0x0)
- `feeder-relais.ota.bin` — app image for **OTA/web update** (Ch. 6.6)
- as a **GitHub release** (versioned file name, e.g. `feeder-relais-v0.0.1.factory.bin`) in the Releases section of the repository.

Build it yourself: `./firmware/build_images.sh` (compiles and updates `firmware/build/`). Otherwise the raw files are located under `firmware/.esphome/build/feeder-relais/.pioenvs/feeder-relais/`.

**The images contain NO Wi-Fi credentials.** Setup runs after flashing via the setup hotspot (Ch. 6.2). This way no foreign Wi-Fi ends up in the delivered product, and the configured data survives OTA updates (fixed storage key).

### Route A — with ESPHome (recommended, one command)

A `secrets.yaml` is **not** needed — no Wi-Fi data is compiled in.

1. Plug in the ESP via **USB-C**.
2. Run in the project folder: `esphome run firmware/timer-relais-c3.yaml` ESPHome compiles, asks for the **serial port** (Linux e.g. `/dev/ttyACM0`, Windows a `COMx`) and flashes.
3. After the restart the ESP opens the setup hotspot → continue with Ch. 6.2.

**Linux tip:** On "Permission denied" for `/dev/ttyACM0`, add your user to the group `dialout`:  
`sudo usermod -aG dialout $USER` (log in again).

### Route B — in the browser, without installation (esptool-js)

For the ready-made `firmware.factory.bin` entirely without ESPHome:

## Ablauf - Erstes Flashen ueber den Browser (esptool-js)

- 1 Chrome oder Edge oeffnen**  
espressif.github.io/esptool-js
- 2 ESP per USB-C anstecken**  
USB-C-Datenkabel verwenden
- 3 "Connect" -> Port waehlen**  
ESP32-C3 als serieller Port
- 4 Flash Address 0 + factory.bin**  
Datei waehlen, dann "Program"
- 5 WLAN einrichten**  
AP "Feeder-Relais Setup" -> feeder-relais.local

1. In **Chrome/Edge** open the page <https://espressif.github.io/esptool-js/>.
2. Plug in the ESP via a **USB-C data cable**.
3. Leave the baud rate at 115200, click **Connect** and select the ESP's serial port in the window (often called "USB JTAG/serial debug unit" or a COMx).
4. Enter 0 at **Flash Address**, use **Choose File** to select the `firmware.factory.bin`, then **Program**. Wait until "Hard resetting..." appears.
5. Continue with the Wi-Fi setup (Ch. 6.2).

If "Connect" does not work: force **boot mode** — hold **BOOT**, briefly tap **RESET**, release **BOOT**, then "Connect" again.

### 6.2 After flashing: set up Wi-Fi

The firmware **deliberately brings no** Wi-Fi credentials with it — after flashing the ESP starts its own **setup hotspot**:

1. On your phone/PC connect to the Wi-Fi **"Feeder-Relais Setup"** (password `feeder1234`).
2. A **captive portal** opens (otherwise open <http://192.168.4.1>).
3. Select your home Wi-Fi, enter the password, save — the ESP restarts and connects.
4. From now on reachable at <http://feeder-relais.local>.

#### Persistence & factory reset:

- The entered Wi-Fi data is stored under a **fixed storage key** and is **preserved across OTA/web updates** (Ch. 6.6) — set up once, then never again.
- For a **factory reset** (deleting the old Wi-Fi data), flash the **factory image with flash erase**: tick "Erase all flash" in the web flasher, or run `esptool.py erase_flash` before writing. A pure OTA update does **not** delete the data.



## 6.3 Troubleshooting when flashing

| Symptom                     | Cause / solution                                                                                                         |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------|
| ESP is not detected at all  | <b>charge cable</b> instead of data cable → different USB-C cable, different USB port                                    |
| No port in the browser      | use Chrome/Edge (WebSerial); on Windows possibly the <b>CH340/CP210x</b> driver; a C3 with native USB usually needs none |
| "Connect" fails             | <b>boot mode</b> : hold BOOT + tap RESET + release BOOT                                                                  |
| Linux "Permission denied"   | user in group dialout (sudo usermod -aG dialout \$USER)                                                                  |
| No ...local after the flash | first set up Wi-Fi via the setup hotspot (Ch. 6.2); mDNS takes a moment                                                  |

After the first flash all updates go **wirelessly** (Ch. 6.6) — the USB socket may well be poorly accessible in the installed state.

## 6.4 Web operation and JSON API (port 80)

`http://feeder-relais.local` opens a mobile web app with five tabs:

- **Start**: three large buttons (trigger T1/T2/T3 with their configured time), live countdown and **Stop**.
- **Times (settings)**: at the very top the **language selection** (Deutsch/English/Français), below it the three timer times (1–600 s), stored persistently.
- **Network (configurable)**: Wi-Fi (SSID/password), IP mode **DHCP/static** (+ IP/gateway/mask/DNS), **NTP server**, a **hostname** (default `feeder-relais`), a switch for **Wi-Fi roaming (802.11k/v)** and a **restart** button.
- **Status**: firmware, uptime, free memory, Wi-Fi, SSID with **channel · colored signal bar · signal** (dBm; 1 bar red, 2 yellow, 3–4 green), **Wi-Fi roaming (802.11k/v) ON/OFF**, IP/MAC, reset reason, relay.
- **Service**: live **log** (display level ERROR/WARN/INFO/DEBUG selectable, can be enabled), **firmware update** (.bin upload, Ch. 6.6) and **restart**.

**Language (multilingual interface)**: The web app and the OLED speak **German, English and French**. The selection is **at the very top of the Times tab** and applies **device-wide** (web and OLED), stored in `g_netcfg.lang` (persistent, default German). The web texts are stored as a JS dictionary in the page (switching without reload); language-dependent status values (Wi-Fi state, reset reason) come from the firmware as neutral codes and are translated in the browser. The OLED translates weekdays, the status (Idle/Feed) and the menu titles (deliberately without accents because of the display's character set). The manual itself is additionally available in [English](#) and [French](#) (the German version is authoritative).

In the **header** are the **hostname** and a **status dot**: green = all is well, yellow = a timer is running, red = fault in the idle state (e.g. OLED not reachable or no Wi-Fi). The **same traffic light** is shown on the device by the dimmed **onboard RGB LED** (WS2812 on GPIO8, see Ch. 5.2).

Technically, `firmware/timer_web.h` (with `firmware/net_config.h` for the persistent network config) serves this page as its own `AsyncWebHandler` on `web_server_base` (the native ESPHome web UI is disabled). The same **JSON API** is also used for integration (e.g. ioBroker) — parameters as a query string, method GET **or** POST:

| Endpoint                                                                       | Effect                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GET /api/status                                                                | JSON with all values: active, remaining, relay, last, times[3], host, ip, ssid, rssi, mac, ap, fw, uptime, heap, wifi, reset, lang, roaming (wifi/reset are language-neutral codes that the web JS translates) |
| POST /api/trigger?button=N                                                     | trigger button N (1-3) (uses its configured time)                                                                                                                                                              |
| POST /api/trigger?seconds=N                                                    | switch on ad hoc for N seconds                                                                                                                                                                                 |
| POST /api/stop                                                                 | switch off immediately                                                                                                                                                                                         |
| POST /api/config?time1=A&time2=B&time3=C                                       | set times (each 1-600 s, persistent) — fields can also be set individually                                                                                                                                     |
| GET /api/net                                                                   | read network config: static, ip, gw, sn, dns, ntp, hostname, roaming, lang                                                                                                                                     |
| POST /api/net?static=0 1&ip=&gw=&sn=&dns=&ntp=&host=&roaming=0 1&lang=de en fr | save network config (fields individually, persistent)                                                                                                                                                          |
| POST /api/wifi?ssid=&pw=                                                       | set Wi-Fi credentials (reconnects)                                                                                                                                                                             |
| POST /api/reconnect                                                            | cleanly reconnect Wi-Fi (applies e.g. changed roaming immediately)                                                                                                                                             |
| POST /api/reboot                                                               | restart the device                                                                                                                                                                                             |
| GET /api/log?level=N&since=M                                                   | log ring buffer as JSON (lines with level ≤ N, seq > M); level 1=ERROR...5=DEBUG                                                                                                                               |

**Applying the network config:** The values are stored in the ESPHome preferences (flash). The **NTP server** takes effect at the next sync (`esp_sntp_setservername`). The **static IP** is set at `wifi.on_connect` via the ESP-IDF `netif` and is therefore active after a **restart** (DHCP is the default). **Wi-Fi** uses ESPHome's own `save_wifi_sta()`; initial access always via the captive portal. The **hostname** is switched at runtime: mDNS (`...local`) immediately via `mdns_hostname_set()`; the DHCP name (in the router) by a **restart of the DHCP client** (`esp_netif_dhcpc_stop/start` → fresh DISCOVER with option 12), i.e. also without a reboot. The name also appears in the **header** of the web app and in the browser tab. It is reduced to a valid DNS label (a-z, 0-9, "-").

**Wi-Fi roaming (802.11k/v):** The capability is compiled in permanently (`enable_btm/enable_rrm` in the YAML → `CONFIG_WPA_11KV_SUPPORT` in the `wpa_supplicant`); the **on/off** is toggled at runtime by the web switch via `g_netcfg.roaming` (persistent). On sets in the STA config the bits 802.11v **BTM** (`set_btm`, the AP/router can specifically re-book the device onto the stronger AP) and 802.11k **RRM** (`set_rrm`, neighbor AP lists) and turns off ESPHome's own scan roaming (the driver takes over); off does the reverse (`set_post_connect_roaming(true)`). Only sensible with **multiple access points with the same SSID** (UniFi/mesh) and when these support 802.11k/v. The bits only go into the STA config at the **next (re)connect**; for that there is the button **"Reconnect now"** in the same card (`POST /api/reconnect` → `wifi.disable + enable`, which briefly disconnects the Wi-Fi), otherwise it takes effect at the next restart. Default: **off**. The checkbox **continuously mirrors the device's real state** (from `/api/status`) and therefore stays visibly ticked after reconnecting; the setting is written to flash **immediately and durably** (`netcfg_save()` calls `global_preferences->sync()`, since ESPHome's `save()` on the ESP32 otherwise only queues it in RAM). The current state is also shown as a row **"Wi-Fi roaming (802.11k/v): ON/OFF"** on the **Status** page.

Example ioBroker (set the time for button 1 to 8 s): `POST http://feeder-relais.local/api/config?time1=8`.

## 6.5 Operation via the three buttons and OLED

The buttons are labeled on the enclosure: **S1 = Down/Manual**, **S2 = SET**, **S3 = UP** (GPIO3/4/5).

### Normal state:

- **short S1 / S2 / S3** → triggers timer 1 / 2 / 3 with the configured time (switches the Shelly on until the time expires).
- **long UP (S3) ≥ 1.2 s** → stops all timers and switches the Shelly **off**.
- **long SET (S2) ≥ 3 s** → opens the **info menu**.

**Info menu** (view only — configuration is done via the web app):

- **S1** pages forward, **S3** backward; **short SET** or **10 s without a press** closes the menu again.
- Pages: 1) Wi-Fi (SSID + signal) · 2) IP address · 3) time + NTP · 4) system (firmware + uptime).

### OLED display (128 × 32):

- **Top:** Wi-Fi signal bars (from RSSI) on the left, the large **time of day** (HH:MM) or, while a timer is running, the **countdown** (e.g. "10 s") in the center, status **Idle/Feed** on the right; in the menu instead the page counter (e.g. "2/4").
- **Bottom:** on the left the **date** (e.g. "Do 16.07.2026"), on the right the **free memory** (kB).

The time base comes via NTP; until the first synchronization the clock shows "--:--". Time zone `Europe/Berlin`. Weekday, status (Idle/Feed) and the menu titles follow the **configured language** (Ch. 6.4).

## 6.6 Firmware updates (OTA)

After the first USB flash, **two wireless update routes** are set up:

- **Network OTA (ESPHome):** `esphome run firmware/timer-relais-c3.yaml` updates over Wi-Fi (port 3232) — USB no longer needed. Configured via `ota: platform: esphome`.
- **Web upload:** In the **Service** tab, upload the compiled `firmware.bin` (or directly `POST /update` on port 80). Configured via `ota: platform: web_server` — runs over `web_server_base`, the native ESPHome web UI is **not** needed for it. The device restarts automatically after the update.

The `firmware.bin` is created by `esphome compile ...` and is located under `.esphome/build/feeder-relais/.pioenvs/feeder-relais/firmware.bin`. The captive portal can also install updates.

## 6.7 Service log / debugging

The **Service** tab shows the last log lines (ring buffer, 40 lines) filtered by the selected level (ERROR/WARN/INFO/DEBUG) when "live display" is enabled. Technique: `logger: on_message:` writes each line into `firmware/log_ring.h`; the web endpoint `GET /api/log?level=&since=` delivers them as JSON. The logger runs at `level: DEBUG` (for VERBOSE, raise the level in the YAML). Complete logs additionally via `esphome logs ...` (UART/network).

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## 7. KiCad — step by step (also for beginners)

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Tested against KiCad 8/9/10; menu names may differ slightly.

## 7.1 Create a project and import the schematic

1. Start KiCad → **File** → **New Project** → name `timer_ersatzplatine`, folder e.g. `hardware/kicad/`.
2. KiCad creates an empty `timer_ersatzplatine.kicad_sch`. **Replace this file in the explorer with our file** (same name!): copy in `hardware/timer_ersatzplatine.kicad_sch`.
3. Open the project, start the schematic editor. KiCad may convert the file to the current format — that is normal, just save.
4. Run **Tools** → **Electrical Rules Check (ERC)**. Warnings about "unconnected power pins" are known and non-critical (our symbols deliberately have no power-pin types); optionally silence them with `PWR_FLAG` symbols on +5V and GND.

## 7.2 Check/assign footprints

**Tools** → **Assign Footprints**. Pre-assigned are:

| Ref              | Footprint                                                  | Action                                                         |
|------------------|------------------------------------------------------------|----------------------------------------------------------------|
| SW1-3            | Button_Switch_THT:SW_PUSH_6mm                              | check (pin pattern 6.9 × 4.4)                                  |
| PS1              | Converter_ACDC:Converter_ACDC_HiLink_HLK-PMxx              | <b>check against the TSP-05 measurement!</b>                   |
| U2               | Package_S0:S0P-4_4.4x2.6mm_P1.27mm                         | for the DIP variant (AQY216 DIP):<br>Package_DIP:DIP-4_W7.62mm |
| F1               | Fuse:Fuse_5x20mm_Horizontal_ReferenceFuseHolder            | or the chosen holder                                           |
| RV1              | Varistor:RV_Disc_D12mm_W3.9mm_P7.5mm                       | S14K275 fits                                                   |
| R1,<br>C1,<br>C2 | standard THT                                               | ok                                                             |
| J1-J3            | pin header/terminal                                        | J2 alternatively as 4 large solder pads                        |
| J4               | Connector_PinSocket_2.54mm:PinSocket_1x04_P2.54mm_Vertical | <b>socket</b> header!                                          |
| U1               | — (empty)                                                  | <b>create yourself</b> , see 7.3                               |

## 7.3 Create a footprint for the ESP32-C3 Super Mini

1. Open the footprint editor → new library `timer_project.pretty` (in the project folder, table "project-specific").
2. New footprint `ESP32-C3_SuperMini`: two pad rows 1×8, pitch 2.54, row spacing 15.24 mm (= 6 × 2.54), pads ø1.7/hole 1.0. Outline 18 × 24 mm on F.Fab, mark the USB side.
3. Assign pad numbers according to the pinout image (left row 5V, G, 3.3, 4, 3, 2, 1, 0 / right row 5, 6, 7, 8, 9, 10, 20, 21) and assign them in the schematic to symbol U1 (the symbol pin names 5V/G/3V3/3/4/5/6/8/9 must point to the correct pad numbers).

## 7.4 Generate the PCB from the schematic

Schematic editor → **Tools** → **Update PCB from Schematic**. All footprints land as a heap next to the (still empty) board.

## 7.5 Import the contour and references

1. PCB editor → **File** → **Import** → **Graphics** → `hardware/platine_original_geometrie.dxf`.

2. The unit is automatically set to **mm** thanks to the DXF header; import first onto layer **User.Drawings**, position (0,0), scale 1.
3. **Dimension check:** measure T1→T2 with the measuring tool — it must give **25.6 mm**. If it is correct, continue; otherwise check the import settings.
4. Select the outer contour (**one closed line** — the 6 edge-open boss slots are already part of the contour, no longer separate circles) → Properties → change layer to **Edge.Cuts**. Everything else (button crosses, mounting areas, window, keepouts, texts) stays on user layers; optionally copy the mounting area frames including their labels additionally onto **F.Silkscreen** (helps with placement).
5. Set the 4 mounting holes as footprints: MountingHole:MountingHole\_3.2mm exactly on the green crosses (position via **E** → type in the coordinates from Ch. 4.3).

## 7.6 Place and route

1. **Buttons first:** SW1–SW3 exactly on (13.2/12.3), (38.8/12.2), (64.1/12.0) — enter the position numerically, do not drag.
2. J4 on the pin crosses (first pin = GND at x 15.0 / y 27.84).
3. PS1/TSP on the mounting-area rectangle (AC pins toward the left edge), U1/ESP on its area (USB toward the top edge), J2 pads at the edge of the Shelly mounting area, J1 + F1 + RV1 in the 230 V corner at the bottom left, U2/PhotoMOS between the ESP area and the Shelly pads, R1 next to it, C1/C2 on the 5 V line near U1.
4. Create **net classes** (File → Board Setup → Net Classes): - HV (nets L\_IN, L\_F, N, SW\_SHELLY, O\_LAST): trace  $\geq 0.8$  mm (the supply path of the Shelly output O  $\geq 2$  mm at 8 A load!), clearance within HV  $\geq 1.0$  mm. - Standard (low voltage): 0.25/0.2 mm.
5. **The golden rule:** between every HV net and every low-voltage net  $\geq 6.0$  mm distance — most easily via a custom rule (Board Setup → Custom Rules): (version 1) (rule HV\_zu\_LV (condition "A.NetClass == 'HV' && B.NetClass != 'HV'") (constraint clearance (min 6.0mm))) The only permitted "approach" is the PhotoMOS itself — its housing is the separation point (pins 1/2 = LV, pins 3/4 = HV).
6. No ground planes/zones in the HV area; no foreign traces under the TSP module.
7. Get **DRC** (Design Rules Check) free of errors.

## 7.7 Manufacturing data and ordering

1. **File → Fabrication Outputs → Gerbers** (standard layers) + drill files.
2. 3D view (Alt+3) as a final visual inspection.
3. Manufacturer (JLCPCB/PCBWay/Aisler): 2 layers, 1.6 mm FR4, lead-free HASL or ENIG, color does not matter. The outer contour including the 6 edge-open boss slots comes automatically from Edge.Cuts.
4. **Before submitting:** print the board 1:1 on paper (File → Plot → PDF, scale 1:1), cut it out and test-fit it in the enclosure!

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## 8. Assembly and commissioning

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1. **Place the small parts** (order flat → tall): R1, C2, U2, buttons (exactly!), J4 socket header, C1, F1 holder, RV1, terminals.
2. **Prepare the ESP32-C3:** flash via USB (Ch. 6), test the Wi-Fi connection, then solder it onto the board (pin headers or directly).

3. **USB-only test:** power the ESP via USB (no mains!) — the buttons must start a countdown, the OLED plugged onto J4 must display, GPIO6 must switch (multimeter/LED on the PhotoMOS input).
4. **Solder in the TSP-05,** glue the Shelly onto the mounting area with VHB, four wires to J2 (L, N, SW, O), a wire to the load output.
5. **Visual inspection + continuity check:** no short between L/N/PE,  $\geq 6$  mm distances maintained, no solder splashes.
6. **First mains test:** enclosure closed, switch on via PRCD/RCD. The OLED shows clock/status → set up the Shelly via button/app (Wi-Fi, Input Mode "Switch") → button test with load.
7. Set the times as desired via <http://feeder-relais.local>.

## 9. File overview

| File                                                   | Content                                                                                | Regenerable?                                                     |
|--------------------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------|
| hardware/platine_template.py                           | <b>source of truth</b> for all geometry; parametric                                    | source                                                           |
| hardware/<br>platine_original_geometrie.dxf            | DXF generated from it for KiCad                                                        | yes: python3<br>platine_template.py (needs<br>pip install ezdxf) |
| hardware/platine_vorschau.png                          | render of the DXF                                                                      | yes (render script see<br>CLAUDE.md)                             |
| hardware/<br>timer_ersatzplatine.kicad_sch             | complete schematic (KiCad 8+)                                                          | hand-maintained                                                  |
| firmware/timer-relais-c3.yaml                          | ESPHome configuration                                                                  | hand-maintained                                                  |
| firmware/timer_web.h                                   | mobile web app + JSON API<br>(C++ handler on<br>web_server_base)                       | hand-maintained                                                  |
| firmware/net_config.h                                  | persistent network config (IP<br>mode, static IP, NTP,<br>hostname, 802.11k/v roaming) | hand-maintained                                                  |
| firmware/log_ring.h                                    | log/debug ring buffer for the<br>Service tab (/api/log)                                | hand-maintained                                                  |
| firmware/build/*.bin                                   | ready-made flash images<br>(factory + ota), current state                              | generated                                                        |
| firmware/build_images.sh                               | compiles and updates<br>firmware/build/                                                | source                                                           |
| docs/DOKUMENTATION.md                                  | this document (German<br><b>source</b> , authoritative)                                | hand-maintained                                                  |
| docs/DOKUMENTATION.en.md, docs/<br>DOKUMENTATION.fr.md | manual in English/French<br>(translation, updated per<br>release)                      | hand-maintained                                                  |
| docs/PROJEKTPLAN.md                                    | phases, status, checklists<br>(German only)                                            | hand-maintained                                                  |
| CLAUDE.md                                              | working instructions for Claude<br>Code                                                | hand-maintained                                                  |
| docs/DOKUMENTATION.pdf, docs/<br>PROJEKTPLAN.pdf       | PDF versions ( <b>mandatory<br/>with every change</b> )                                | yes: python3 tools/md2pdf.py<br>docs/*.md                        |
| tools/md2pdf.py                                        |                                                                                        | source                                                           |

| File           | Content                                  | Regenerable?    |
|----------------|------------------------------------------|-----------------|
|                | Markdown→PDF generator (embeds images)   |                 |
| docs/img/*.png | diagrams of the flashing guide (Ch. 6.1) | generated (PIL) |

## 10. Change history of the geometry template

| Version | Change                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v1      | first coordinates from perspective-corrected photos ( $\pm 1\text{--}2\text{ mm}$ )                                                                                                                                                                                                                                                                                                                                        |
| v2      | scanner measurement: $76.7 \times 66.1$ , buttons/holes/cutout precise (solder-side scan)                                                                                                                                                                                                                                                                                                                                  |
| v3      | enclosure fit $105.3 \times 79.4$ , 6 boss cutouts $\varnothing 10$ , old cutout omitted (= B5)                                                                                                                                                                                                                                                                                                                            |
| v4      | correction: no recessing possible (only 1.8 mm below) — window removed, Shelly on the mounting area at the right, 4th hole back, rib keepouts                                                                                                                                                                                                                                                                              |
| v5      | TSP-05 planned instead of HLK (bottom left), ESP to the top left                                                                                                                                                                                                                                                                                                                                                           |
| v6      | OLED frame-mounted: window area freed up, J4 connection                                                                                                                                                                                                                                                                                                                                                                    |
| v7      | labeled placeholders (DXF texts)                                                                                                                                                                                                                                                                                                                                                                                           |
| v8      | OLED plug-in mounting: J4 constructed vertically on the window centering                                                                                                                                                                                                                                                                                                                                                   |
| v9      | boss positions measured directly (Silvio 13.07.: pitch 45, row spacing 70, upper line 6 mm above the top edge, left column 6 mm beside the left edge $\rightarrow x -6/39/84, y -6/+64$ ); 6 boss cutouts changed from $\varnothing 10$ holes to <b>edge-open U-slots</b> (more tolerant during insertion); corner radii at the outer slots locally reduced to 3.2/2.1; contour remains a closed Edge.Cuts loop (verified) |